

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Debugging & Optimization Task

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	Total Marks:	100
CO2 Statement:	Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.		
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Section / Batch:	SLOT-B	Date:	22/07/2026

Code of Conduct

I am GAUTHAM G (192524371) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: GAUTHAM G

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure	List Operations	6 – 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
GRAND TOTAL:		100 Marks	

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, well-documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well-justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

OPEN

✓ Completed

Review →

Debugging & Optimization Task

10/10 answered

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

OPEN

✓ Completed

Review →

Debugging & Optimization Task

10/10 answered

CO1_AT1_C CODE OUTPUT PREDICTION

OPEN

✓ Completed

Review →

C CODE OUTPUT PREDICTION

10/10 answered

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Q6

Linked List
5 marks

Q7

Stack
5 marks

Q8

Linked List Data Structures
5 marks

Q9

Linked List
5 marks

Q10

Queue
5 marks

10/10 answered

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int binarySearch(int arr[],int n,int key){
2     int low=0,high=n-1;
3     while(low<=high){
4         int mid=(low+high)/2;
5         if(arr[mid]==key)
6             return mid;
7         else if(arr[mid]<key)
8             low=mid+1;
9         else
10            high=mid-1;
11
12    }
13    return -1;
14 }
```

Submitted

Test Input

5
10 20 30 40 50
40

Output

```
Compilation Error:
C:/CCPPComp/CCPP1/bin/./lib/gcc/
x86_64-w64-
mingw32/10.3.0/./././././x86_64
-w64-mingw32/bin/ld.exe:
C:/CCPPComp/CCPP1/bin/./lib/gcc/
```

Pending manual review

Q2 Stack

5 marks

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List
5 marks

Find the error in the stack deletion operation. Justify it.

```
int pop() {
    if(top == 0) {
        printf("Underflow");
        return -1;
    }
    return stack[top--];
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int pop() {
2     if(top == -1) {
3         printf("underflow");
4         return -1;
5     }
6     return stack[top--];
7 }
```

Test Input

```
4
5 10 15 20
15
```

Output

CODE_712_DEBUGGING & OPTIMIZATION TASK (SET 1)

QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack
5 marks

Q6

Q3 Stack

5 marks

Check the snippet and identify the logical error.

```
int stack[5];
int top;
void init() {
    top = 0;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int stack[5];
2 int top;
3 void init() {
4     top=-1;
5 }
```

Test Input

```
stdin input (optional)
```

Output

QUESTIONS

Q1

Binary Search
5 marks

Q2

Stack
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Stack
5 marks

Q6

Linked List
5 marks

Q4 Stack

5 marks

Identify the errors in the following snippet.

```
if(ch == ' '){  
while(stack[top] != ' '){  
postfix[j++] = pop();  
}  
pop();  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 if(ch==' '){  
2     while(stack[top]!=' '){  
3         postfix[j++] = pop();  
4     }  
5     pop();  
6 }
```

Test Input

stdin input (optional)

Output

QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List
5 marks

Q5 Stack

5 marks

In balancing Bracket, you identify the error.

```
if((ch == ']' && stack[top] == '[') ||
(ch == '}' && stack[top] == '{')) {
    pop();
} else {
    return 0;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 if((ch=='&&stack[top]=='[') ||
2 ((ch=='&&stach[top]=='{')) {
3     pop();
4 }else {
5     return 0;
6 }
```

Test Input

stdin input (optional)

Output

QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List
5 marks

Q6 Linked List

5 marks

What is the issue?

```
temp = head
while temp.next != NULL:
    print(temp.data)
    temp = temp.next
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 temp=head;
2 while(temp!=null) {
3     printf("%d",temp->data);
4     temp=temp->next;
5 }
```

Test Input

stdin input (optional)

Output

QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List
5 marks

Q7 Stack

5 marks

Point out the error.
POP(stack):
if top == 0:
print("Underflow")
else:
return stack[top--]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 POP(stack):
2 if(top==0):
3     print("underflow")
4     else
5         return stack[top--]
```

Test Input

stdin input (optional)

Output

QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List
5 marks

Q8 Linked List Data Structures

5 marks

Identify the errors in the following code
while current != NULL:
current.next = prev
prev = current
current = current.next

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 while(current!=null){
2     next=current->next;
3     ccurrent->next=prev;
4     current=next;
5 }
```

Test Input

stdin input (optional)

Output

Q9 Linked List

5 marks

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List
5 marks

What is the error in the below code?

```
new.next = head
head = new
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 newnode->next=head;
2 head=newnode;
```

Test Input

stdin input (optional)

Output

matslab.saveetha.com/practice_app/dashboard_student

QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List
5 marks

Q10 Queue

5 marks

What is missing in the below pseudocode?

```
DEQUEUE(Q):
if front == -1:
    print("Underflow")
else:
    return Q[front++]
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 DEQUEUE(Q):
2 if(front==-1)
3     printf("underflow");
4 else{
5     item=Q[front++];
6     if(front>rear)
7         front=rear=-1;
8     return item;
9 }
```

Test Input

stdin input (optional)

Output

- Q3 Binary Search 5 marks
- Q4 Queue 5 marks
- Q5 Queue 5 marks
- Q6 Doubly Linked List 5 marks
- Q7 Stack 5 marks
- Q8 Stack 5 marks
- Q9

```
// perform operation: result = op1 + op2;
push(result);
}
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 while(expression[i]!='\0'){
2     if(isdigit(expression[i]))
3         push(expression[i]-'0');
4     else{
5         int op2=pop();
6         int op1=pop();
7         int result=op1+op2;
8         push(result);
9     }
10    i++;
11 }
```

Test Input

stdin input (optional)

Output

```
Compilation Error:
C:\Windows\TEMP\sandbox_192524371
RB_6a72f0bae369f3.30524975\main.c
:1:1: error: expected identifier
or '(' before 'while'
1 |
```

- 5 marks
- Q2 Circular Queue 5 marks
- Q3 Binary Search 5 marks
- Q4 Queue 5 marks
- Q5 Queue 5 marks
- Q6 Doubly Linked List 5 marks
- Q7 Stack 5 marks

```
int dequeue() {
if (front == -1) return -1;
int data = queue[front];
front = (front + 1) % MAX;
return data;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int dequeue() {
2     if(front==--1)
3         return -1;
4     int data=queue[front];
5     if(front==rear) {
6         front=rear=-1;
7     }
8     else{
9         front=(front+1)%MAX;
10    }
11    return data;
12 }
```

Test Input

stdin input (optional)

Output

Q1
Stack Notations - Postfix
Expression
5 marks

Q2
Circular Queue
5 marks

Q3
Binary Search
5 marks

Q4
Queue
5 marks

Q5
Queue
5 marks

Q6
Doubly Linked List
5 marks

Q7

Identify the bug in this Recursive Binary Search.

```
int binarySearch(int arr[], int l, int r, int x) {  
    int mid = l + (r - l) / 2;  
    if (arr[mid] == x) return mid;  
    if (arr[mid] > x) return binarySearch(arr, l, mid, x);  
    return binarySearch(arr, mid, r, x);  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int binarySearch(int arr[], int l, int r, int x) {  
2     if (l > r)  
3         return -1;  
4     int mid = l + (r - l) / 2;  
5     if (arr[mid] == x)  
6         return mid;  
7     if (arr[mid] > x)  
8         return binarySearch(arr, l, mid - 1, x);  
9     return binarySearch(arr, mid + 1, r, x);  
10 }
```

Test Input

stdin input (optional)

Output

QUESTIONS

Q1
Stack Notations - Postfix
Expression
5 marks

Q2
Circular Queue
5 marks

Q3
Binary Search
5 marks

Q4
Queue
5 marks

Q5
Queue
5 marks

Q6
Doubly Linked List
5 marks

Q4 Queue

5 marks

Find the error in this Deque insertFront operation.

```
void insertFront(int data) {  
    if (front == 0) front = MAX - 1;  
    else front = front - 1;  
    deque[front] = data;  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 void insertFront(int data) {  
2     if (front == 0)  
3         front = MAX - 1;  
4     else  
5         front = front - 1;  
6     deque[front] = data;  
7 }
```

Test Input

stdin input (optional)

Output

QUESTIONS

Q1
Stack Notations - Postfix Expression
5 marks

Q2
Circular Queue
5 marks

Q3
Binary Search
5 marks

Q4
Queue
5 marks

Q5
Queue
5 marks

Q6
Doubly Linked List

Q5 Queue

5 marks

Identify the error in the Queue enqueue logic.

```
void enqueue(int data) {
    if (rear == MAX) return;
    queue[rear++] = data;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 void enqueue(int data) {
2     if (rear==MAX-1)
3         return;
4     queue[++rear]=data;
5 }
```

Test Input

stdin input (optional)

Output

Q6 Doubly Linked List

5 marks

Identify the error in this Doubly Linked List node deletion.

```
void deleteNode(Node* del) {
    if (del->prev != NULL)
        del->prev->next = del->next;
    if (del->next != NULL)
        del->next->prev = del->prev;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 void deleteNode(Node*del) {
2     if (del==NULL)
3         return;
4     if (del->prev!=NULL)
5         del->prev->next=del->next;
6     if (del->next!=NULL)
7         del->next->prev=del->prev;
8     free(del);
}
```

Test Input

stdin input (optional)

Output

Q1
Stack Notations - Postfix Expression
5 marks

Q2
Circular Queue
5 marks

Q3
Binary Search
5 marks

Q4
Queue
5 marks

Q5
Queue
5 marks

Q6
Doubly Linked List
5 marks

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1

Stack Notations - Postfix
Expression
5 marks

Q2

Circular Queue
5 marks

Q3

Binary Search
5 marks

Q4

Queue
5 marks

Q5

Queue
5 marks

Q7 Stack

5 marks

Identify the issue in the below pseudocode.
POP():
if top == 0 → Underflow

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 POP() :  
2 if (top == -1)  
3     underflow;  
4 else  
5     top--;
```

Test Input

stdin input (optional)

Output

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1

Stack Notations - Postfix
Expression
5 marks

Q2

Circular Queue
5 marks

Q3

Binary Search
5 marks

Q4

Queue
5 marks

Q5

Queue

Q8 Stack

5 marks

Point out the error.
if top == MAX:
Overflow

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 if (top == MAX-1)  
2     Overflow;
```

Test Input

stdin input (optional)

Output

QUESTIONS

Q1
Stack Notations - Postfix
Expression
5 marks**Q2**
Circular Queue
5 marks**Q3**
Binary Search
5 marks**Q4**
Queue
5 marks**Q5**
Queue
5 marks**Q6**
Doubly Linked List**Q9 Circular Queue**

5 marks

Identify the issue in the following snippet for Circular Queue in Traffic Signal.

```
rear = (rear + 1) % MAX;  
queue[rear] = vehicle;  
if(rear == front){  
    printf("Queue Full");  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 if((rear+1)%MAX==front){  
2     printf("queue full");  
3     return;  
4 }  
5 rear=(rear+1)%MAX;  
6 queue[rear]=vehicle;  
7
```

Test Input

stdin input (optional)

Output

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

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QUESTIONS

Q1
Stack Notations - Postfix
Expression
5 marks**Q2**
Circular Queue
5 marks**Q3**
Binary Search
5 marks**Q4**
Queue
5 marks**Q5**
Queue
5 marks**Q10 Stack**

5 marks

Point out the error?
PEEK(stack):
return stack[top+1]**Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int PEEK() {  
2     if(top==--1)  
3         return -1;  
4     return stack[top];  
5 }
```

Test Input

stdin input (optional)

Output